Fundamentals 17

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1. We do this by showing that f is Lipschitz continuous, so we want to show that there exists K such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in \mathbb{R}$. Since $f'(x)$ is bounded, let $A \leq f'(x) \leq B$ for all x, and let $K = \max(|A|, |B|)$. We claim that $|f(x) - f(y)| \leq K|x - y|$ for all x, y.

Assume not, for contradiction. Then there exist some $x, y \in \mathbb{R}$ such that $\frac{|f(x)-f(y)|}{|x-y|} > K$. This means the slope between the points $(x, f(x))$ and $(y, f(y)$ is $> K$. By MVT, there exists $a \in [x, y]$ such that $f'(a) > K = \max(|A|, |B|)$, but $A \leq f'(x) \leq B$, so we have a contradiction.

2a.

$$\frac{d}{dx}(kf) = \lim_{h\to 0} \frac{kf(x+h) - kf(x)}{h} = k \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = k\frac{df}{dx}.$$

2b.

$$\frac{d}{dx}(f+g) = \lim_{h\to 0} \frac{f(x+h) - f(x) + g(x+h) - g(x)}{h} = \frac{df}{dx} + \frac{dg}{dx}.$$

3. $(x^2 \sin x)' = (x^2)' \sin x + x^2(\sin x)' = 2x \sin x + x^2 \cos x$.

4. By chain rule, $(\sin(x^n))' = \cos(x^n)(x^n)' = nx^{n-1} \cos(x^n)$.

5. We have that

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$= \lim_{h\to 0} \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h\to 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h\to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{\sqrt{x} + \sqrt{x}} = \frac{1}{2\sqrt{x}}.$$

6. We claim that

$$(f_1 f_2 \dots f_n)' = (f_1' f_2 \dots f_n) + (f_1 f_2' f_3 \dots f_n) + \dots + (f_1 \dots f_{n-1} f_n').$$

We can prove this by induction. From the product rule, this is true when $n = 2$. Now, assume that the above is true. Then

$$(f_1 f_2 \dots f_n f_{n+1})' = (f_1 f_2 \dots f_n)' f_{n+1} + (f_1 f_2 \dots f_n) f'_{n+1}$$

$$= (f'_1 f_2 \dots f_n) + (f_1 f'_2 f_3 \dots f_n) + \dots + (f_1 \dots f_{n-1} f'_n) + (f_1 f_2 \dots f_n f'_{n+1}),$$

as desired.

7.

$$f(x) = \tan x, \quad f(x)\cos x = \sin x,$$

$$(f(x)\cos x)' = \cos x, \quad f'(x)\cos x - f(x)\sin x = \cos x,$$

$$f'(x) = \frac{\cos x + (\tan x)(\sin x)}{\cos x} = 1 + \tan^2 x = \sec^2 x.$$

8.

$$f(x) = \sec x, \quad f(x)\cos x = 1,$$

$$(f(x)\cos x)' = 0, \quad f'(x)\cos x - f(x)\sin x = 0,$$

$$f'(x) = \frac{(\sec x)(\sin x)}{\cos x} = (\sec x)(\tan x).$$

9. Let $f : \mathbb{R} \to \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{otherwise.} \end{cases}$$

The derivative is $2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$, which is discontinuous at 0.

We claim that $f'(0)$ exists, and we can prove this using squeeze theorem. We have that

$$f'(0) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{f(h)}{h} = \lim_{h\to 0} h \sin\left(\frac{1}{h}\right).$$

Since $-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|$ for all $x \neq 0$ and $\lim_{x\to 0} |x| = \lim_{x\to 0} -|x| = 0$, by squeeze theorem, $\lim_{h\to 0} h \sin\left(\frac{1}{h}\right) = 0$, so $f'(0) = 0$.